Non-Functional Requirements (NFRs)

AI-Powered Predictive Maintenance System

Purpose

Define the quality attributes and operational constraints that ensure the system is reliable, secure, usable, explainable, and suitable for predictive maintenance

NFR Overview

Non-Functional Requirements define **how well the system should perform**, rather than what functions it provides

The system shall satisfy the following quality requirements

.**Performance** — Fast processing and response time

.**Accuracy & Reliability** — Reliable predictions with controlled false alarms

.**Explainability** — Understandable AI/ML predictions

.**Usability** — Simple and role-appropriate interface

.**Security** — Protected access and secure system resources

.**Data Privacy** — Protection of machine and user data

.**Availability** — System remains accessible during operation

.**Maintainability** — Easy monitoring, updating, and retraining

.**Scalability** — Ability to support additional machines and data

.**Auditability** — Traceable predictions, alerts, and user actions

.**Cost Control** — Controlled cloud and LLM usage

.**Safety** — AI recommendations must not directly control machinery

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Performance Requirements

NFR-PERF-01 — Prediction Response Time

The system shall return AI/ML inference results within **3 seconds** under normal operating conditions

NFR-PERF-02 — Dashboard Response

The dashboard shall load the main fleet overview within **5 seconds** under normal system load

NFR-PERF-03 — Alert Update

The system shall update machine status and alert information within **5 seconds** after successful processing of new sensor data

NFR-PERF-04 — Report Generation

The system shall generate a standard machine diagnostic PDF report within **10 seconds**, excluding external LLM response time

Related Requirements: FR-AI-10, FR-UI-01, FR-REP-01

Accuracy & Reliability Requirements

NFR-ACC-01 — Prediction Quality

The deployed model shall meet the minimum performance thresholds defined during model evaluation before production deployment

NFR-ACC-02 — False Alarm Control

The system shall monitor false alarms and missed failures using historical or validation data

NFR-ACC-03 — RUL Reliability

RUL predictions shall be accompanied by an uncertainty/confidence range to communicate prediction uncertainty

NFR-ACC-04 — Model Validation

No AI/ML model shall be deployed unless its evaluation results have been recorded and .compared with the current model

NFR-ACC-05 — Data Quality

Invalid, corrupted, missing, or out-of-range sensor records shall not be used directly for model inference

Related: FR-AI-01, FR-AI-05, FR-AI-06, FR-AI-11, FR-AI-12

Explainability Requirements

NFR-XAI-01 — Prediction Explanation

Each critical or warning prediction shall provide an explanation showing the main sensor features contributing to the result

NFR-XAI-02 — Explanation Availability

The system shall provide SHAP/LIME explanations for supported deployed models

NFR-XAI-03 — Understandable Explanations

Explanations shall be presented in a form understandable to Reliability Engineers and Maintenance Technicians

NFR-XAI-04 — Prediction Traceability

Each explanation shall be associated with its corresponding prediction and model version

Related: FR-AI-08, FR-UI-03, FR-AI-13, FR-AI-16

Example

Critical Condition

- High Vibration
- Elevated Temperature
- Abnormal Pressure

Model: LSTM-v2.1

Usability Requirements

NFR-USE-01 — Role-Based Interface

The system shall provide views appropriate to the user's role and responsibilities

NFR-USE-02 — Simple Status Representation

Machine health shall be clearly represented using the defined severity levels

Normal / Warning / Critical

NFR-USE-03 — Alert Actions

Operators shall be able to Acknowledge or Escalate an alert using a clear and simple interface

NFR-USE-04 — Responsive Interface

The dashboard shall be usable on common desktop and tablet screen sizes

NFR-USE-05 — Clear Navigation

Main system functions shall be accessible through clearly labeled navigation elements

Related: FR-UI-01, FR-UI-05

Security & Data Privacy Requirements

Security

NFR-SEC-01 — Authentication

Only authenticated users shall access protected system functions

NFR-SEC-02 — Role-Based Access Control

Users shall only access features and data permitted by their assigned role

NFR-SEC-03 — API Key Protection

LLM/API keys shall never be displayed or stored in plain text

NFR-SEC-04 — Secure Configuration

System configuration and sensitive credentials shall be protected from unauthorized access

Data Privacy

NFR-PRIV-01 — Data Minimization

Only the data required for an LLM request shall be sent to an external provider

NFR-PRIV-02 — Local-Only Mode

The system shall support a local LLM mode in which LLM requests remain within the plant environment

NFR-PRIV-03 — External Data Protection

Sensitive plant information shall not be unnecessarily exposed to external services

Related: SH-09, FR-LLM-04, UC-2

Availability, Maintainability & Scalability

Availability

NFR-AVL-01
The system shall remain available during normal plant operating periods

NFR-AVL-02
If the cloud LLM service becomes unavailable, the system shall use the configured local fallback when available

Maintainability

NFR-MNT-01
.AI/ML models shall be versioned so that previous versions can be identified and restored

NFR-MNT-02
Model experiments, parameters, datasets and evaluation metrics shall be traceable

NFR-MNT-03
The system shall support model retraining when data drift or degraded performance is detected

Scalability

NFR-SCL-01
The system architecture shall allow additional machines and sensor streams to be added without redesigning the complete application

NFR-SCL-02
The system shall support increasing sensor data volume while maintaining the defined performance targets

Related: FR-AI-13, FR-AI-14, FR-AI-15, FR-LLM-04

Auditability, Cost Control & Safety

Auditability

NFR-AUD-01
The system shall log important prediction events including machine ID, timestamp, model version, severity and prediction results

NFR-AUD-02
Alert status changes shall record the responsible user and timestamp

NFR-AUD-03
Threshold changes shall record the old value, new value, user and timestamp

Cost Control

NFR-COST-01
The system shall minimize unnecessary external LLM API calls

NFR-COST-02
The system shall support a local LLM fallback to reduce dependency on cloud API usage

NFR-COST-03
LLM and cloud usage shall be monitored to support cost control

Safety

NFR-SAFE-01
The system shall operate as a decision-support system only and shall not directly control machinery

NFR-SAFE-02
The LLM shall refuse requests to bypass safety procedures or directly control equipment

NFR-SAFE-03
Critical alerts shall be clearly distinguished from lower-severity conditions

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NFR Traceability & Summary

Category	NFR IDs	Main Related Requirements
Performance	NFR-PERF-01–04	FR-AI-10, FR-UI-01, FR-REP-01
Accuracy & Reliability	NFR-ACC-01–05	FR-AI-01, 05, 06, 11, 12
Explainability	NFR-XAI-01–04	FR-AI-08, FR-UI-03
Usability	NFR-USE-01–05	FR-UI-01, FR-UI-05
Security	NFR-SEC-01–04	FR-LLM-04, UC-21
Data Privacy	NFR-PRIV-01–03	FR-LLM-04
Availability	NFR-AVL-01–02	FR-LLM-04
Maintainability	NFR-MNT-01–03	FR-AI-13–15
Scalability	NFR-SCL-01–02	AI/ML Architecture
Auditability	NFR-AUD-01–03	FR-AI-16, UC-12
Cost Control	NFR-COST-01–03	FR-LLM-04, FR-REP-03
Safety	NFR-SAFE-01–03	FR-LLM-03, FR-AI-07

Final Goal

The NFRs ensure that the predictive maintenance system is

Fast • Reliable • Explainable • Usable • Secure • Private • Maintainable • Scalable • Auditable • Safe